

ThinkWell — Backend Deployment & Cloud Infrastructure

Deployed and maintained the backend infrastructure for **ThinkWell**, a native mobile wellness/therapy application designed to help users track their physique and overall wellbeing.

What I worked on

- Deployed the **Node.js/TypeScript backend** to an AWS EC2 server.
- Configured **Node.js 22**, production dependencies, environment variables, and the application runtime.
- Set up **PM2** for process management, automatic restarts, and keeping the backend running reliably.
- Configured **Nginx** as a reverse proxy to route domain traffic to the backend service.
- Configured **SSL/HTTPS** and resolved production certificate/domain issues.
- Set up and connected **Redis** running locally on the server for backend functionality.
- Worked with the application's **MongoDB/Mongoose** data layer and production configuration.
- Used **Postman/API documentation** to test and verify backend endpoints.
- Integrated **New Relic** for application monitoring and troubleshooting.
- Configured an **AWS Elastic IP** so the server's public address remained stable.
- Diagnosed and fixed production **build, deployment, connectivity, and SSL-related issues**.

Problems I solved

The main challenge was turning a development backend into a stable production service that could be accessed reliably by the mobile application.

I handled issues involving the production build, server configuration, process management, reverse proxying, HTTPS/SSL, domain connectivity, Redis connectivity, and server IP changes. I also used logs, API testing, and monitoring to identify and troubleshoot runtime problems instead of relying only on local development behavior.

Result

The backend was deployed as a production service with a stable public endpoint, HTTPS, process supervision through PM2, Nginx reverse proxying, Redis support, monitoring, and the infrastructure required for the mobile application to communicate with the backend reliably.